ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Statistical inference of transcriptomics data
Part II: comparing methods

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Comparing methods

- Methods validation often done in multiple steps.
- Simulations:
  - pro - ground truth known;
  - pro - simulate diverse scenarios: robustness/sensitivity analyses (vary number of samples, cells, sparsity data, strength of difference we aim to detect, etc...);
  - cons - only partially realistic: real data generation process usually more complex (biology + measurement).
- Real data analyses:
  - pro - realistic;
  - cons - lack of ground truth: build it in lab (expensive, and limited: only for a few genes; not always possible), or lists of "reasonable results" (based on current biological knowledge): e.g., `https://www.proteinatlas.org/search/wound+healing`.

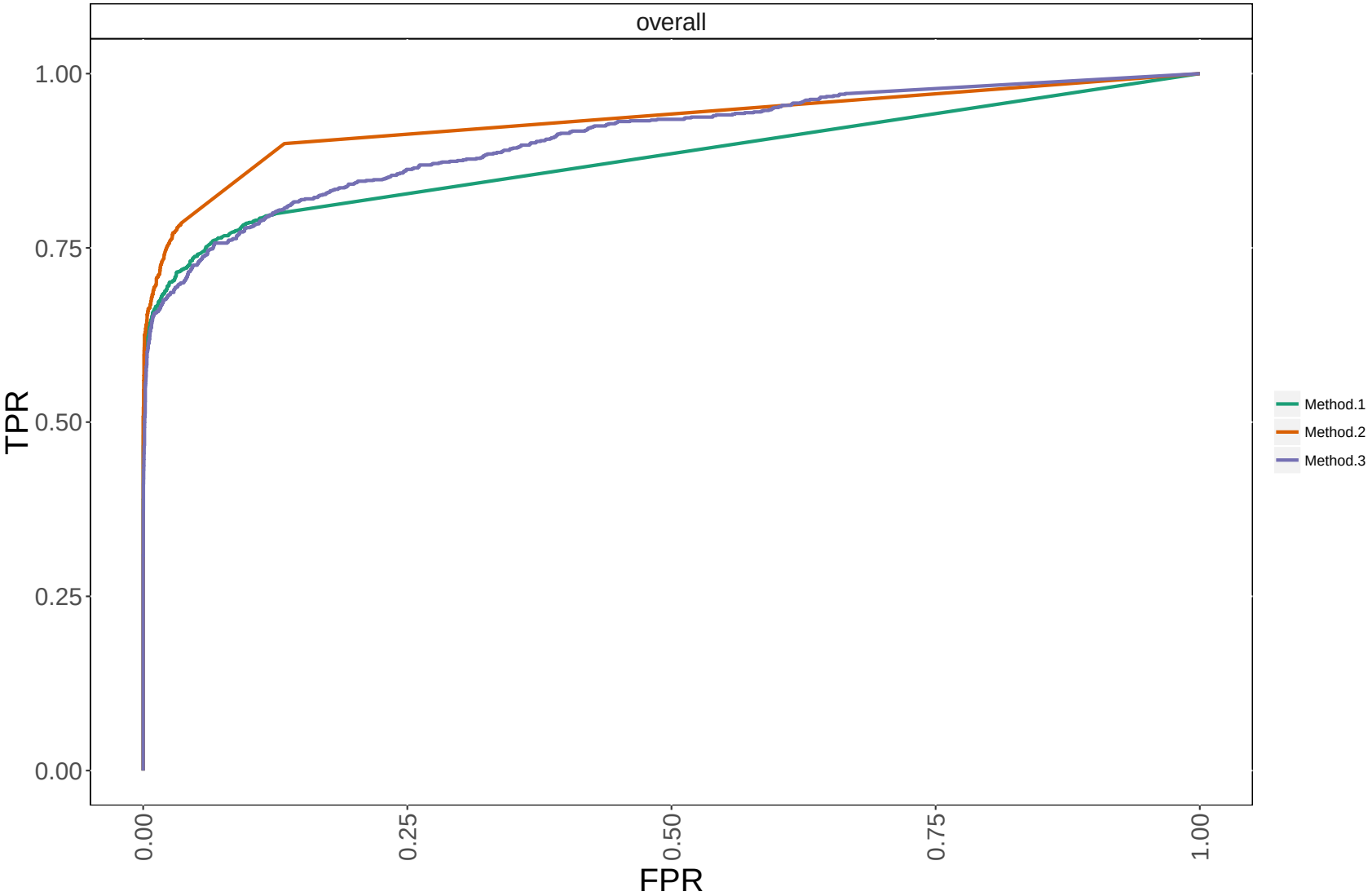
Comparing methods

- Semi-simulated datasets (more realistic simulations):
  - generate a simulation, using real data as anchor;
  - parametric: estimate model parameters from real data -> use them to generate simulations;
  - non-parametric: real data measurements -> re-arrange them.

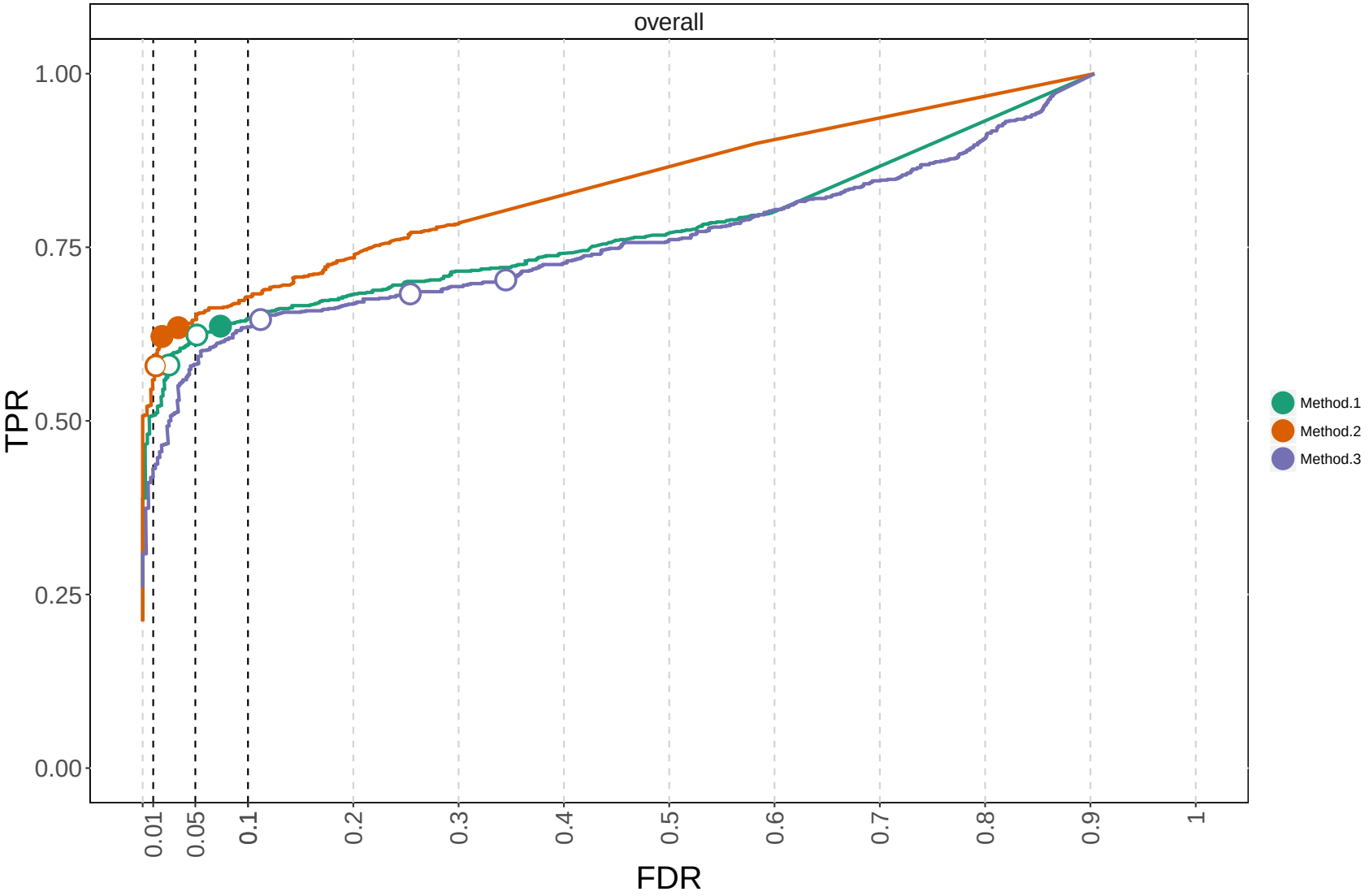
Assessing a method's performance

- Method comparisons based on simulations where we know the ground truth.
- Ground truth typically binary: 0 (e.g., no differential gene/pathway) or 1 (e.g., differential gene/pathway).
- Main plots:
  - **ROC**: true positive rate (TPR) vs false positive rate (FPR); Diagonal line = the blind/random model;
  - **TPR vs FDR** (more interesting than the ROC in this field) - check if FDR is really controlled;
  - **FPs** vs top significant genes: focus on the very top genes.
- In both ROC and FDR plots, focus on the left area - top selected genes.

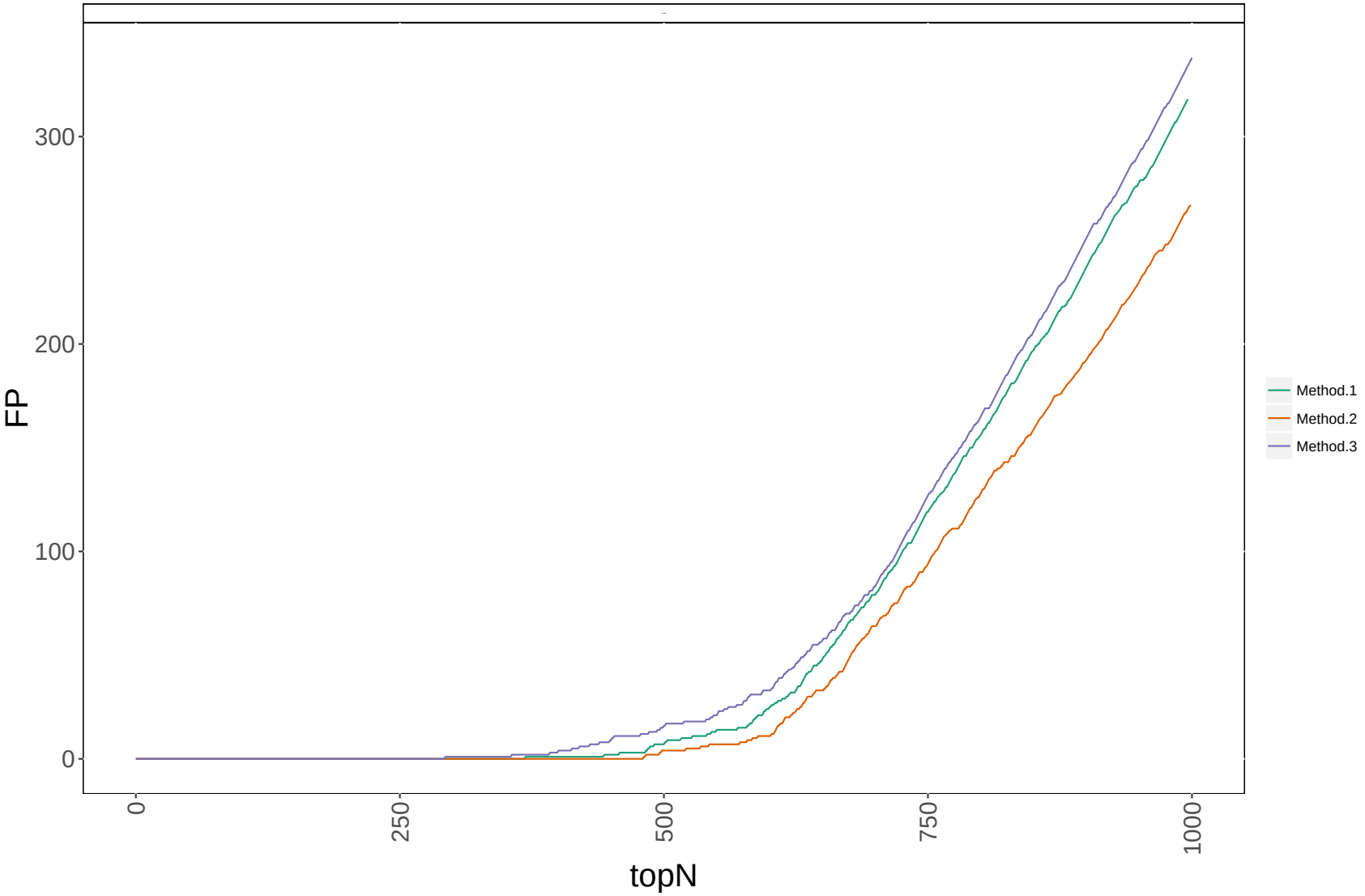
ROC curve


overall
1.00
0.75
0.50
0.25
0.00
TPR
0.00
0.25
0.50
0.75
1.00
FPR
Method.1
Method.2
Method.3

TPR vs FDR plot

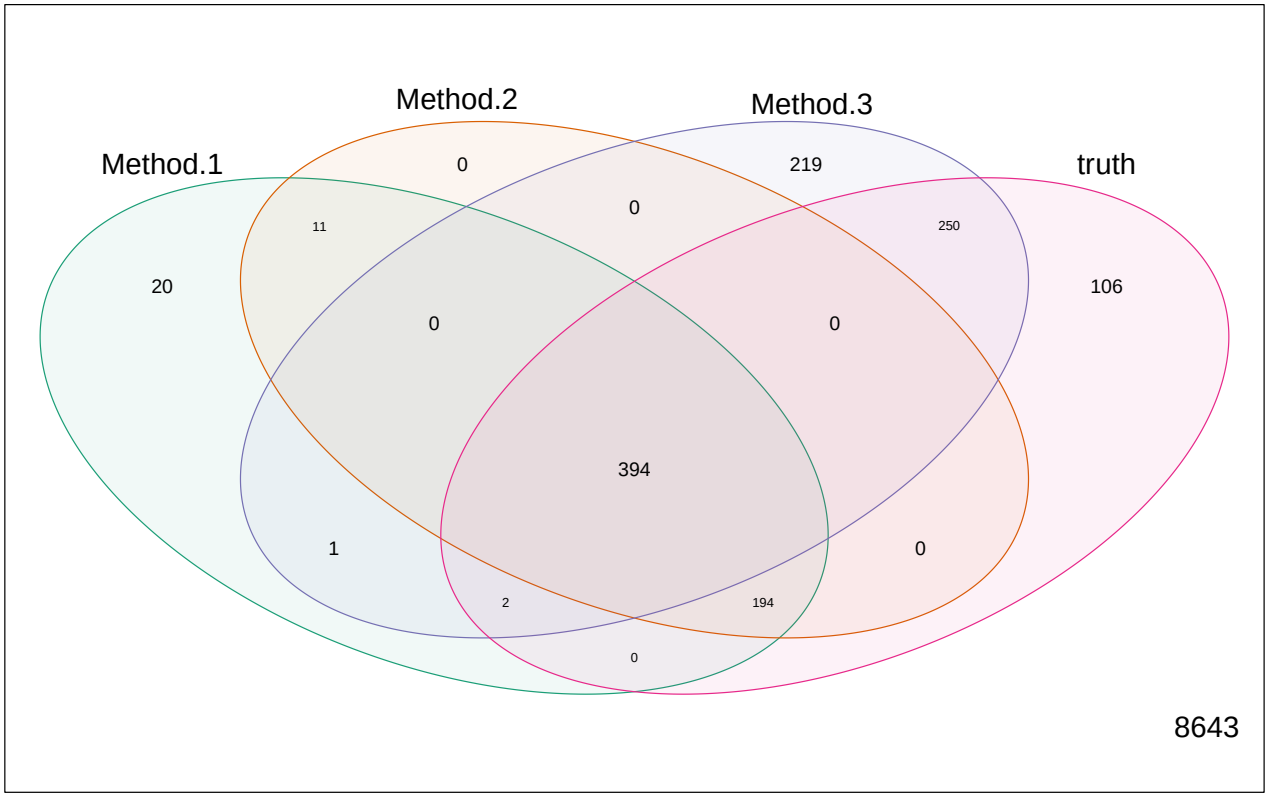


FPs vs top selected genes



Venn diagram

- Even without a ground truth, it shows similarities among methods.
- Set a cutoff and compare significant results for each method.



What comes after RNA-seq studies?

- The main aim of RNA-seq studies is to identify genes/transcripts/pathways which show interesting characteristics (DGE, DAS, ORA, etc...): these genes will typically be studied more in depth in further biological analyses.
- This is why we mostly care about top (most significant) results: we often select a few (very) significant results that undergo further studies.